

3D Organ Shape Completion and Reconstruction from Incomplete or Noisy Medical Data Using Geometric Deep Learning

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1. CWa-T1 Project Title and Abstract

1.1 Project Title

3D Organ Shape Completion and Reconstruction from Incomplete or Noisy Medical Data Using Geometric Deep Learning

1.2 Abstract

Accurate reconstruction of three-dimensional organ shapes from incomplete or noisy medical data remains an important challenge in biomedical engineering. Reliable 3D anatomical models are valuable for applications such as surgical planning, anatomical visualisation, and patient-specific intervention design. However, many existing approaches still rely on volumetric voxel-based representations, which can become computationally expensive at high resolutions and may struggle to preserve fine surface geometry and anatomical detail. This motivates further investigation into whether geometry-aware learning methods can provide a more suitable alternative in medically realistic reconstruction settings.

This project aims to investigate the potential of geometric deep learning (GDL) for 3D organ shape completion and reconstruction under incomplete or corrupted input conditions. Rather than assuming that geometry-aware approaches are universally superior, the project focuses on evaluating whether such methods can offer advantages over a voxel-based baseline in tasks where surface detail preservation, anatomical plausibility, and robustness to missing input are especially important. Based on the current literature, the project adopts a narrower and more controlled scope by focusing on one main anatomical target rather than claiming broad cross-organ generalisation from the outset.

The research will include a structured review of existing literature, the preparation of suitable 3D medical shape data, and the development of a reproducible input corruption pipeline to simulate incomplete or noisy anatomical observations. Within this controlled setting, one volumetric baseline and one geometry-aware reconstruction approach will be compared using consistent evaluation criteria, including reconstruction accuracy, surface quality, robustness to missing data, and computational cost.

The expected outcome is a reproducible research-stage pipeline that supports a clearer methodological comparison between voxel-based and geometry-aware reconstruction strategies, and provides a better-grounded basis for future work on reliable 3D anatomical shape completion.

2. CWa-T2 Planning Review Protocol: Research Questions

2.1 Research Questions

Based on the scope already defined in the abstract, the research questions of this project are framed around the task itself, the key constraints faced by existing methods, and the possible technical solution paths. More specifically, this project focuses on how three-dimensional shape completion and reconstruction can be performed for a target organ under incomplete or noisy medical input conditions (P). At the same time,

particular attention is given to surface detail preservation, anatomical plausibility, robustness to missing or corrupted input, and computational cost at relatively high resolution (C). On this basis, the review will compare the performance of geometry-aware deep learning approaches with volumetric / voxel-based baselines (S). This problem setting is consistent with the aims of the project and also provides a clear framework for the later extraction of studies in Table 6 and the synthesis in CWb-T2.

Within this framework, the review will address the following research questions:

RQ1: What main methodological routes have been proposed in the current literature for three-dimensional organ or anatomical shape completion and reconstruction from incomplete or noisy medical data?

RQ2: Under controlled input corruption conditions, how do geometry-aware deep learning approaches compare with volumetric / voxel-based methods in terms of reconstruction accuracy, surface detail preservation, anatomical plausibility, robustness to missing input, and computational cost?

RQ3: What is the strength of the current evidence supporting these approaches? More specifically, what are the main strengths and limitations of the existing studies in terms of dataset selection, input corruption simulation, evaluation metrics, experimental design, and biomedical relevance?

RQ4: Based on the existing literature, how should this project further refine and narrow its choices regarding target organ selection, data processing pipeline, model comparison strategy, and evaluation protocol design?

2.2 Research Question Development

During the initial literature screening process, two key insights had a direct impact on the assessment of the research questions’ sufficiency.

First, among the 98 de-duplicated papers, a large number of studies (n=57) were excluded at title and abstract screening due to task irrelevance or methods falling outside the intended scope of this project. After full-text eligibility assessment, 18 studies were finally retained for structured comparison in Table 6. The final included literature was dominated by studies on cardiac reconstruction and cranial or skull completion, with additional evidence on abdominal organs, vertebrae, and large intestine shape completion. However, studies demonstrating robust cross-organ generalisation remained very limited. This distribution revealed a significant gap in the existing literature regarding generalisable geometry-aware anatomical shape completion, further strengthening the necessity of this project.

Second, several voxel-based reconstruction studies identified during screening remained useful as comparative baselines, but many of them did not address geometry-aware surface modelling directly. This finding further validated the rationale of comparing geometry-aware methods with voxel-based baselines in RQ2 and supported the core assumption that representation choice is central to reconstruction quality, robustness, and anatomical plausibility.

Based on the screening findings, the four research questions in T2 do not require structural modification. The original formulations remain appropriate after exposure to the literature, and the selected evidence supports the relevance of the defined constraint dimensions. The final research question framework therefore remains well justified.

3. CWb-T2 Background and Critical Evaluation

3.1 Background and Critical Evaluation

This review mainly focuses on the question of how three-dimensional organ shape completion and reconstruction can be performed from incomplete or noisy medical data. According to Table 6, the 18 included studies cover several task areas, including cardiac reconstruction, cranial defect repair, and applications involving abdominal organs, liver and tumour, large intestine, and vertebrae. From the

current distribution of the literature, research on cardiac reconstruction appears to be relatively more systematic. Overall, this field already has a certain research foundation, but the evidence is still uneven across different organs, input conditions, and representation types. In particular, current support remains limited on the question of whether these methods can be stably generalised across different organs.

For 3D organ reconstruction models, one group of studies is more focused on geometry-aware representations, meaning that they model directly on meshes, graphs, or point clouds. The second group is closer to traditional volumetric or voxel-based approaches. For this project, the first group of studies is more important, because they are more closely related to the core issue of how reconstruction can be achieved while preserving surface geometric detail and anatomical plausibility. Representative studies in Table 6 include *Slice2Mesh*, *MR-Net*, *Mesh Deformation U-Nets*, and direct whole-heart mesh reconstruction [3, 10, 13, 15]. Although these papers differ in their specific inputs and task settings, they share one clear feature: they do not treat the 3D surface as only a final post-processing result, but instead model the surface itself as the learning target. The advantage of this design is that when the input is already sparse, or when the goal is a high-quality surface rather than only voxel segmentation, the model is often better able to preserve structural continuity and anatomical shape.

Among these studies, *Slice2Mesh* is a relatively typical image-to-mesh method [3]. It starts directly from sparse cine MRI slices and predicts the left ventricular surface through graph-based deformation. This paper is especially informative for the current project because it suggests that even without full dense 3D supervision, topology-consistent 3D surfaces can still be obtained if template priors and image features are used effectively. In comparison, *MR-Net* and *Mesh Deformation U-Nets* are more oriented toward contour- or point-cloud-to-mesh reconstruction [10, 13]. Both indicate that when the input itself is incomplete, affected by misalignment, or missing contours, template deformation and graph-structured learning can improve reconstruction stability. Another study, direct whole-heart mesh reconstruction, further shows the potential of the direct image-to-mesh route in cardiac tasks, especially in terms of anatomical consistency and temporal consistency [15]. Taken together, these studies suggest that the value of geometric representations is not simply that the results look better, but that in some tasks they make it easier to preserve surface structure, topological continuity, and anatomical plausibility.

However, the current literature does not fully support the simple conclusion that geometry-aware methods are always better than voxel-based methods. *Carver* in Table 6 is a useful counterexample [4]. Although it is not a mesh-native GDL method and instead uses a voxel-based strategy for right ventricle reconstruction, it still performs well in a sparse multi-view ultrasound setting. This suggests that while representation is important, it is not the only factor that determines performance. The way the input is organised, how the task itself is defined, how suitable the network architecture is, and whether the evaluation protocol is appropriate can all have a clear impact on the final outcome. In other words, if the current project later compares geometry-aware methods with voxel baselines, it would not be reasonable to assume in advance that the geometry-aware representation must always win. A more convincing comparison should be made under the same input corruption conditions and with a unified evaluation framework.

Besides cardiac reconstruction, another important line of work is point-cloud completion or generative refinement. In Table 6, *Multi-class PCCN*, *GRFormer*, *CLAP*, and *Shape completion in the dark* can all be placed in this category [1, 2, 6, 7]. Together, they suggest that point-cloud-based methods are often more flexible when the input is only partially visible, sparse, or already affected by errors from upstream segmentation. For example, *Multi-class PCCN* focuses on cardiac anatomy reconstruction from cine MRI. It not only addresses sparse input, but also explicitly handles misalignment, and it includes relatively large-scale external validation, making it one of the stronger pieces of evidence in the current literature [7]. *GRFormer* combines point clouds with Transformer-based design and applies the approach to more complex anatomical structures such as liver and tumour, suggesting that point-cloud completion may be more flexible than fixed-topology mesh methods when the organ boundaries are more complex

and variable [2]. *CLAP* further introduces generative models into shape refinement and highlights that when upstream segmentation contains discontinuities, false positives, or missing regions, shape priors can help pull the output back toward a more anatomically plausible space [1]. As for *Shape completion in the dark*, its value lies in placing the problem in a heavily occluded ultrasound setting and reminding us that in medical tasks, synthetic data generation and corruption simulation may themselves be essential parts of a successful method, rather than only secondary preprocessing steps [6].

At the same time, this line of research also has some clear limitations. First, although many methods achieve good results on their own tasks, the input corruption settings are not uniform. Some studies deal with sparse contours, some with slice misalignment, some with segmentation defects, and some with visibility problems in ultrasound [1, 6, 7, 10, 13]. Although these papers appear to focus on 3D shape completion, they are not actually solving exactly the same problem. Second, the evaluation metrics are not consistent. Different studies use different measures such as Chamfer distance, Hausdorff distance, mean surface distance, Dice, and volume similarity, which means that the results are not always directly comparable across papers. Finally, although a number of methods perform well on a single organ or a single dataset, the evidence becomes much weaker when cross-organ, cross-dataset, or even cross-modality validation is considered. This suggests that the current literature is better at showing that a method is feasible under a particular setting than proving that a certain approach is already broadly suitable for all medical shape completion tasks.

In terms of strength of evidence, the studies in Table 6 that can serve as the main references are still those in the cardiac direction, such as *Slice2Mesh*, *Multi-class PCCN*, *MR-Net*, *Mesh Deformation U-Nets*, and direct whole-heart mesh reconstruction [3, 7, 10, 13, 15]. These studies are not only methodologically more mature, but are also relatively more complete in terms of dataset scale, experimental design, and baseline comparison. By contrast, studies such as *CLAP*, *GRFormer*, and *Shape completion in the dark* are highly informative, but they mainly suggest that new directions are worth paying attention to, rather than already providing strong general evidence [1, 2, 6]. As for background papers or earlier methodological studies, they still have some reference value, but they are more suitable for use in discussing method development or conceptual background, rather than serving as the most central comparative evidence in the main body of the review [16–18].

Based on these studies, the initial idea of this project needs some adjustment. The original plan leaned more toward broadly discussing the general advantages of geometry-aware methods across different organs. However, based on the current literature, the available evidence is not strong enough to support such a broad claim about cross-organ generalisability. A more reasonable direction would be to narrow the project scope to a more controlled setting: first select one main target organ, then build a reproducible simulation pipeline for incomplete or noisy inputs, and finally compare one voxel baseline with one geometry-aware method under a unified evaluation framework. This would be more consistent with the strongest evidence currently available in the literature, and would also be more suitable for developing into a practical and executable experimental design. In this sense, the present review has not overturned the original project direction, but it has helped refine the project from a relatively broad idea into something more testable and more clearly structured as a research project.

3.2 Conclusions

Overall, this literature review suggests that geometric deep learning has clear potential in medical 3D shape completion and reconstruction, especially in tasks where the input is sparse, partially missing, or where preserving surface detail and anatomical plausibility is important. Existing studies have already shown that geometry-aware representations such as meshes, graphs, and point clouds can perform well in a number of specific tasks [3, 7, 10, 13, 15]. This is particularly true in cardiac reconstruction, where a relatively coherent methodological line and stronger experimental support have already emerged. By comparison, studies on other organs are beginning to appear, but they are still more scattered, and both

the task settings and the strength of evidence are less mature than in the cardiac literature [1, 2, 6, 9].

Based on what the current literature supports, a cautious conclusion would be that in tasks involving sparse, incomplete, or surface-sensitive inputs, geometry-aware methods that incorporate anatomical priors may have a better chance of outperforming purely voxel-based baselines in terms of surface quality, structural continuity, and anatomical plausibility, but this advantage is usually task-dependent rather than universally applicable [4, 5, 8, 12]. Therefore, the most suitable next step for this project is not to aim directly for a universal cross-organ model, but to focus first on one target organ and compare one geometry-aware method with one voxel baseline under controlled input corruption settings. The evaluation should also not rely on a single performance metric alone, but should jointly consider reconstruction error, surface plausibility, robustness to missing input, and computational cost.

4. References

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A. Appendix

A.1 Table 1. Sources Documentation Table

Table 1: Table 1. Sources Documentation Table

Data Source	Type	Date of Search	Years Covered by Search	Justification
PubMed	Digital Library / Bibliographic Database	10/04/2026	2015–2026	PubMed provides strong coverage of biomedical and clinical research. It is suitable for identifying studies related to medical 3D anatomical structures, organ modelling, completion, and reconstruction, and it supports the biomedical application context of this project.
IEEE Xplore	Digital Library	10/04/2026	2015–2026	IEEE Xplore provides strong coverage of artificial intelligence, machine learning, computer vision, 3D geometric modelling, and engineering implementation. It is particularly suitable for identifying methodological studies on geometric deep learning, graph neural networks, and mesh-based learning.
Scopus	Abstract and Citation Database	10/04/2026	2015–2026	Scopus is a broad interdisciplinary database covering computer science, engineering, and biomedical research. It supports comprehensive initial searching and helps identify relevant studies across intersecting research areas.
Web of Science Core Collection	Citation Index Database	10/04/2026	2015–2026	Web of Science Core Collection indexes high-quality journals and conference papers. It is useful for complementary searching and citation tracking, helping identify influential studies, representative methods, and core evidence in the field.

A.2 Table 2. Search Strategy Keywords Table

Table 2: Table 2. Search Strategy Keywords Table

Term	Synonymous Groups of Keywords
3D anatomical target	3D organ, organ shape, anatomical shape, anatomical structure, organ mesh, 3D anatomy, cranial implant, skull reconstruction, bone reconstruction, CT, MRI
Shape completion / reconstruction task	shape completion, shape reconstruction, 3D reconstruction, surface reconstruction, mesh completion, shape generation, 3D inpainting
Geometric deep learning method	geometric deep learning, graph neural network, graph convolutional network, mesh-based learning, point cloud deep learning, PointNet, MeshCNN, dynamic graph CNN, volumetric deep learning
Incomplete / noisy input condition	incomplete data, noisy data, partial input, missing regions, corrupted input, sparse data, partial point cloud, partial scan

A.3 Table 3. Search Strategy Table

Table 3: Table 3. Search Strategy Table

ID	Source	Search Query	Justification	Results
1	PubMed	(3D organ OR organ shape OR anatomical shape OR anatomical structure OR organ mesh OR 3D anatomy OR cranial implant OR skull reconstruction OR bone reconstruction OR CT OR MRI) AND (shape completion OR shape reconstruction OR 3D reconstruction OR surface reconstruction OR mesh completion OR shape generation OR 3D inpainting)	Initial broad search within PubMed combining anatomical structure and shape completion terms to establish the overall scope of relevant biomedical literature and demonstrate the narrowing effect of subsequent constraints	70,470
2	PubMed	(3D organ OR organ shape OR anatomical shape OR anatomical structure OR organ mesh OR 3D anatomy OR cranial implant OR skull reconstruction OR bone reconstruction OR CT OR MRI) AND (shape completion OR shape reconstruction OR 3D reconstruction OR surface reconstruction OR mesh completion OR shape generation OR 3D inpainting) AND (geometric deep learning OR graph neural network OR graph convolutional network OR mesh-based learning OR point cloud deep learning OR PointNet OR MeshCNN OR dynamic graph CNN OR volumetric deep learning) AND (incomplete data OR noisy data OR partial input OR missing regions OR corrupted input OR sparse data OR partial point cloud OR partial scan)	Refined search within PubMed adding geometric deep learning method terms and incomplete input condition terms to the anatomical and task terms. PubMed’s dedicated biomedical coverage provides sufficient indexing density to support all term categories without over-restriction	32

ID	Source	Search Query	Justification	Results
3	IEEE Xplore	(3D organ OR organ shape OR anatomical shape OR anatomical structure OR organ mesh OR 3D anatomy OR cranial implant OR skull reconstruction OR bone reconstruction OR CT OR MRI) AND (shape completion OR shape reconstruction OR 3D reconstruction OR surface reconstruction OR mesh completion OR shape generation OR 3D inpainting) AND (geometric deep learning OR graph neural network OR graph convolutional network OR mesh-based learning OR point cloud deep learning OR PointNet OR MeshCNN OR dynamic graph CNN OR volumetric deep learning) AND (incomplete data OR noisy data OR partial input OR missing regions OR corrupted input OR sparse data OR partial point cloud OR partial scan)	Full search within IEEE Xplore combining all term categories. As a dedicated engineering and computer science database with comprehensive coverage of deep learning and geometric modelling literature, all term categories are retained and expected to yield relevant results without over-restriction	34
4	Scopus	TITLE-ABS-KEY(("3D organ" OR "organ shape" OR "anatomical shape" OR "anatomical structure" OR "organ mesh" OR "3D anatomy" OR "cranial implant" OR "skull reconstruction" OR "bone reconstruction") AND ("shape completion" OR "shape reconstruction" OR "3D reconstruction" OR "surface reconstruction" OR "mesh completion" OR "shape generation") AND ("geometric deep learning" OR "graph neural network" OR "graph convolutional network" OR "mesh-based learning" OR "point cloud" OR "PointNet" OR "volumetric deep learning"))	The fourth group was omitted as pilot searches returned only one result with all four groups, indicating that Scopus's interdisciplinary coverage of this specialised topic is insufficient to support the additional constraint, the incomplete input condition is treated as implicit within the shape completion task terms to preserve sufficient retrieval breadth in this interdisciplinary database	36
5	Web of Science	TS=("3D organ" OR "organ shape" OR "anatomical shape" OR "anatomical structure" OR "organ mesh" OR "3D anatomy" OR "cranial implant" OR "skull reconstruction" OR "bone reconstruction") AND TS=("shape completion" OR "shape reconstruction" OR "3D reconstruction" OR "surface reconstruction" OR "mesh completion" OR "shape generation") AND TS=("geometric deep learning" OR "graph neural network" OR "graph convolutional network" OR "mesh-based learning" OR "point cloud" OR "PointNet" OR "volumetric deep learning")	The same methodological rationale as Scopus applies: the incomplete input condition is treated as implicit within the shape completion task terms to preserve sufficient retrieval breadth in this interdisciplinary database	17

A.4 Table 4. Inclusion and Exclusion Criteria

Table 4: Table 4. Inclusion and Exclusion Criteria

Type	Criteria	Justification
Inclusion	Studies proposing or evaluating deep learning methods for 3D anatomical or organ shape completion, reconstruction, or generation	Directly addresses RQ1 and RQ2, targeting studies that propose solutions to the core problem using learning-based approaches
Inclusion	Studies reporting quantitative evaluation using geometric or surface quality metrics such as Chamfer Distance, Hausdorff Distance, or IoU	Addresses RQ3; studies without quantitative results cannot contribute meaningful comparative evidence
Inclusion	Studies published in peer-reviewed journals or conference proceedings between 2018 and 2026	GDL methods matured after 2017 with PointNet; earlier studies are unlikely to reflect current methodological standards
Exclusion	Studies operating exclusively on 2D image representations without any 3D shape component	Outside the 3D mesh-based scope of this project and cannot address RQ1 or RQ2
Exclusion	Studies involving non-anatomical 3D shapes such as furniture or vehicles	Outside the biomedical scope defined by the project’s problem statement
Exclusion	Review or survey articles	Do not provide primary experimental evidence required by RQ2 and RQ3
Exclusion	Studies for which full text is unavailable	Cannot be assessed for quality or relevance

A.5 Table 5. Quality Assessment Criteria

Table 5: Table 5. Quality Assessment Criteria

Quality Criteria	Evaluation Questions
Study Design	Is the research objective clearly stated? Is the choice of 3D shape representation (mesh, point cloud, voxel grid) explicitly justified in relation to the task? Is the overall experimental design sufficient to support the reported conclusions?
Dataset	Is the dataset publicly available and reproducible? Does it contain 3D anatomical or organ shapes relevant to medical applications? Is the dataset size sufficient to support meaningful training and evaluation of deep learning models?
Input Corruption Simulation	Is the method for simulating incomplete or noisy input explicitly described? Is the corruption process reproducible? Does the study evaluate performance across varying levels of incompleteness or noise?
Model and Method	Are the architectural design decisions of the proposed model justified? Is the implementation sufficiently described to allow reproducibility? Is the model compared against at least one relevant baseline method?
Evaluation Metrics	Are the evaluation metrics appropriate for assessing 3D shape quality, such as Chamfer Distance, Hausdorff Distance, or surface-to-surface distance? Are the metrics clearly defined and consistently applied?
Results and Evidence	Are the quantitative results thoroughly analysed and discussed? Does the reported evidence support the conclusions drawn? Are limitations of the method acknowledged?
Ethics	This project does not involve human participants, clinical data collection, or patient-identifiable information, as it operates solely on publicly available 3D shape datasets. Ethical considerations are therefore not applicable as a quality filter in this review, though studies utilising patient-derived imaging data should confirm appropriate ethical approval.
Generalisability	Is the method evaluated on more than one organ type or dataset? Do the authors discuss the applicability of their findings beyond the specific experimental setting?

A.6 Figure 1. Venn Diagram

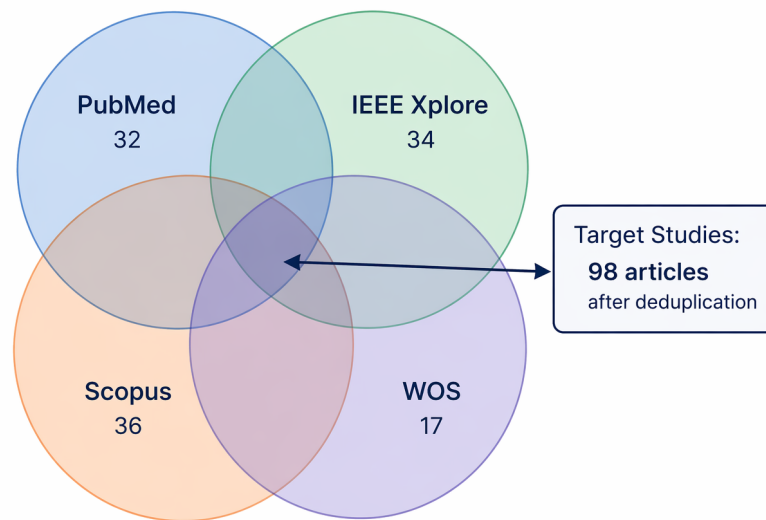


Figure 1: Venn diagram of search results and target studies across four databases, with the intersection representing the 98 deduplicated articles entering the selection process.

A.7 Figure 2. Study Selection Flow Chart

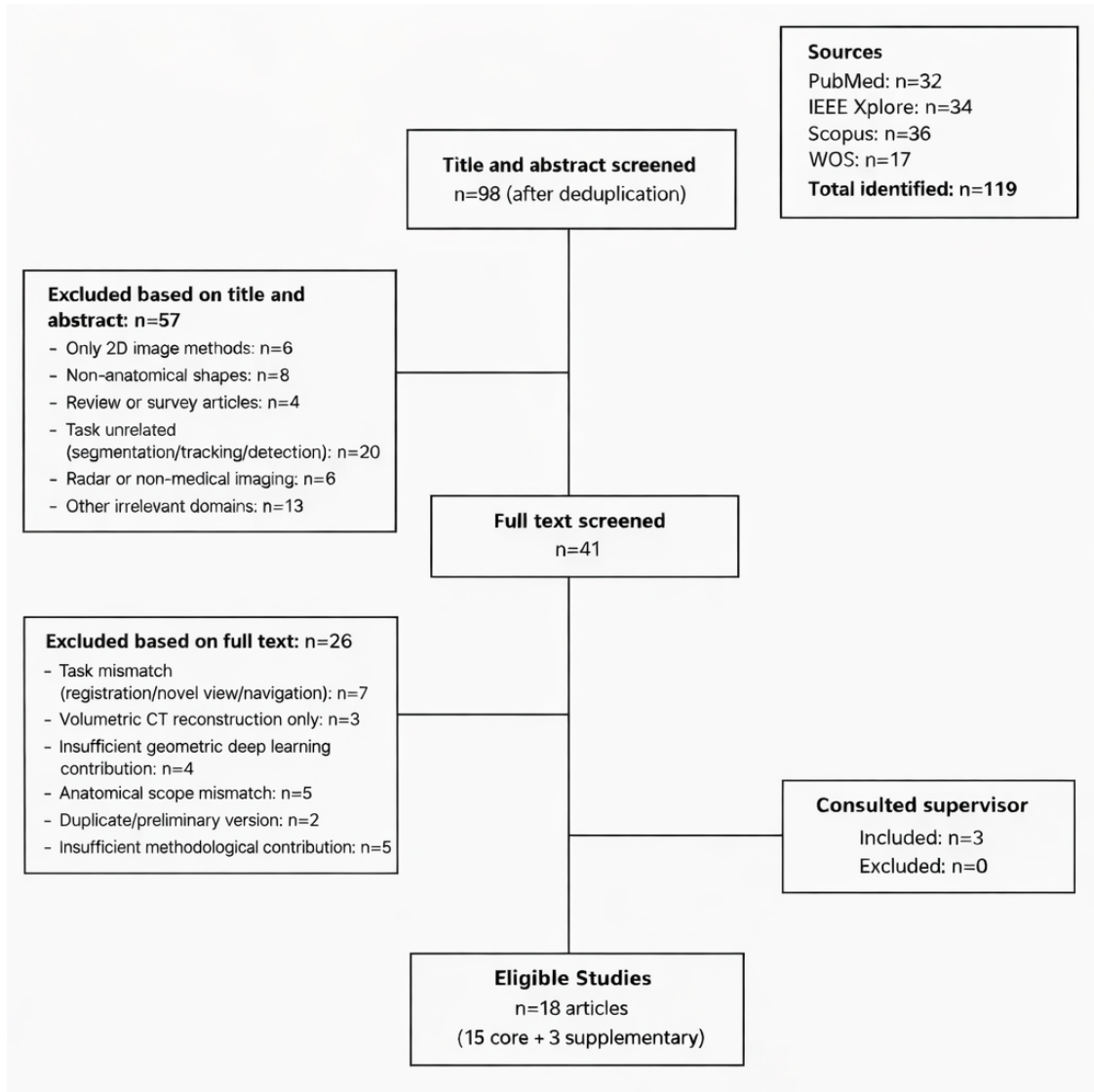


Figure 2: Study selection and quality screening flow chart summarising the complete literature search and screening process.

Table 6: Table 6. Data extraction and evidence assessment of the 18 eligible studies retained for CWb synthesis.

Study ID / Ref	Characteristics	Data	Methods	Results	Conclusions	AI/ML Eval.
S01 [1] Mouheb et al. (2026) <i>CLAP</i>	3D organ shape refinement for large intestine after poor segmentation.	TotalSegmentator + Duke CT; train/val/test split; additional real defective cases.	Point-cloud sampling; hierarchical VAE + latent diffusion refinement + mesh reconstruction.	Improved over initial shapes and strong baselines; real-case improvement reported.	Generative refinement can recover plausible bowel geometry, but still needs post-processing.	Perf: high; Gen: moderate-high; Int: low-mod; Scalability: moderate.
S02 [2] Wang et al. (2026) <i>GRFormer</i>	3D reconstruction of complex abdominal anatomy (liver/tumour) from sparse contour-driven completion.	LiTS and SDPH datasets; clinical contour-derived sparse inputs.	Dual-branch feature extractor (3D CNN + Transformer) with multi-stage point generation.	Strong relative performance for liver/tumour completion under sparse inputs.	Transformer-based point completion is effective for highly irregular anatomy, but output remains point-cloud based.	Perf: high; Gen: moderate; Int: low; Scalability: moderate.

Continued on next page

Study ID / Ref	Characteristics	Data	Methods	Results	Conclusions	AI/ML Eval.
S03 [3] Xiao et al. (2025) <i>Slice2Mesh</i>	image-to-mesh reconstruction for left ventricle from sparse cine MRI slices under partial supervision.	Clinical cine MRI; 150 test samples for mesh reconstruction; 99 subjects for downstream MI classification.	2D UNet feature sampling + iterative GCN-based mesh deformation from a template; ARAP regularisation.	Outperformed traditional baselines; downstream 4D mesh use clinically promising.	Strong recent image-to-mesh method with topology-consistent LV meshes and limited supervision.	Perf: high; Gen: moderate; Int: low-mod; Scalability: good.
S04 [4] Li et al. (2025) <i>Carver</i>	Core; RV reconstruction from sparse multi-view echocardiography; voxel route.	1,278 instances from 32 patients; patient-level cross-validation.	Dual-channel voxel input (convex hull + contours) + 3D U-shaped CNN with attention.	Strong quantitative performance and robustness to sparse views.	Excellent cardiac benchmark, but not mesh-native GDL.	Perf: high; Gen: moderate; Int: low-mod; Scalability: good.
S05 [5] Mazzocchetti et al. (2024) <i>Occupancy Networks</i>	Core; cranial implant / skull completion; implicit-geometry comparison study.	SkullBreak, SkullFix, plus qualitative real craniotomy testing.	Conditioned convolutional occupancy network from partial point cloud to full skull occupancy.	Competitive quality with lower memory than dense 3D CNNs.	Shows the value of implicit representation for high-resolution completion.	Perf: high; Gen: moderate; Int: low; Scalability: high.
S06 [6] Gafencu et al. (2024) <i>Shape completion in the dark</i>	Core; vertebra completion from heavily occluded ultrasound views.	VerSe20 synthetic generation; paired US/CT patients; phantom data.	US-aware synthetic data generation + probabilistic point-cloud completion + surface reconstruction.	Worked on synthetic and patient data; anatomy-specific landmarks preserved.	Physics-based synthetic training is important for ultrasound completion.	Perf: mod-high; Gen: promising but small real set; Int: moderate; Scalability: moderate.
S07 [7] Beetz et al. (2023) <i>Multi-class PCCN</i>	Core; fully automatic cardiac reconstruction from cine MRI using point completion.	Synthetic development data + 1,000 UK Biobank cases.	MRI segmentation to sparse point clouds, then multi-class point completion and mesh conversion.	Improved over benchmark baselines; strong large-scale external validation.	Strong point-cloud cardiac pipeline with explicit handling of sparsity and misalignment.	Perf: high; Gen: high; Int: low; Scalability: good.
S08 [8] Wodzinski et al. (2023) <i>PC completion transformers</i>	Core; cranial defect reconstruction with transformer-based point completion.	SkullBreak and SkullFix challenge datasets.	Iterative PoinTr-style completion with low-memory point-cloud processing.	Near-volumetric accuracy with much lower memory usage.	Useful evidence for efficiency advantages of point-cloud completion.	Perf: mod-high; Gen: good; Int: low; Scalability: high.
S09 [9] Wang et al. (2023) <i>Abdominal GCN</i>	Core; abdominal organ completion from partially visible regions.	Standardised abdominal organ meshes; liver highlighted.	GCN mesh completion with topology enhancement.	Topology-aware design improved completion accuracy.	Useful for discussing graph structure and partial-observation completion.	Perf: moderate; Gen: unclear; Int: moderate; Scalability: moderate.
S10 [10] Beetz et al. (2022) <i>Mesh Deformation U-Nets</i>	Core; mesh-native cardiac reconstruction from misaligned MRI contours.	Synthetic SSM data + UK Biobank application.	Template fitting + spectral graph convolutions + mesh pooling/up-sampling.	Outperformed smoothing and GCN baselines; realistic population-level reconstructions.	Strong example of mesh-native GDL for misalignment correction.	Perf: high; Gen: good; Int: moderate; Scalability: good.
S11 [11] Hu et al. (2022) <i>SRT</i>	Core in retained set, but weaker; brain point cloud from single 2D MRI.	In-house brain MRI dataset.	CNN-free transformer with attention-based point-cloud generation.	Better than comparison methods on its own dataset.	Broadens method coverage, but evidence is narrower and less clinically persuasive.	Perf: moderate; Gen: limited; Int: low; Scalability: good.

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Study ID / Ref	Characteristics	Data	Methods	Results	Conclusions	AI/ML Eval.
S12 [12] Li et al. (2021) <i>Skull defect restoration</i>	Core; volumetric medical shape completion for cranial defect repair.	Synthetic skull defects with limited real-case testing.	Volumetric CNN completion with patch-based training; implant by subtraction.	Strong restoration quality reported; useful classic volumetric baseline.	Demonstrates the strengths and cost of dense volumetric completion.	Perf: mod-high; Gen: limited; Int: low; Scalability: moderate.
S13 [13] Chen et al. (2021) <i>MR-Net</i>	Core; learned deformation for cardiac reconstruction from sparse/incomplete point clouds.	Cardiac contours / point clouds under incomplete and automatic settings.	Hybrid GCN template-deformation network.	Accurate reconstruction with robustness to incomplete contours.	Important precursor for sparse contour-to-mesh cardiac reconstruction.	Perf: high; Gen: high; Int: moderate; Scalability: good.
S14 [14] Beetz et al. (2021) <i>Biventricular PCN</i>	Core; early biventricular reconstruction from cine MRI contours.	SSM-derived datasets + small UK Biobank test.	Point completion network from sparse contours to dense point cloud; meshing afterward.	Strong synthetic performance and very fast inference.	Important early study, though later work provides fuller validation.	Perf: high; Gen: moderate; Int: low; Scalability: high.
S15 [15] Kong et al. (2021) <i>Direct whole-heart mesh</i>	Core; direct whole-heart mesh reconstruction from cardiac images.	Multiple CT/MR cardiac settings, including temporal evaluation.	Direct image-to-mesh graph-convolutional template deformation.	Strong accuracy and good anatomical / temporal consistency reported.	Key reference for direct mesh reconstruction in cardiac imaging.	Perf: very high; Gen: good; Int: moderate; Scalability: good.
S16 [16] Heidari & Iosifidis (2025) <i>CAD survey</i>	Supplementary background; survey of GDL for CAD rather than medical imaging.	Survey of CAD datasets and representations.	Review of mesh / point / graph / B-Rep learning methods.	No unified experiment; used for conceptual taxonomy only.	Useful for representation background, not core empirical evidence.	Background only; not directly comparable as primary evidence.
S17 [17] Fu et al. (2023) <i>GDL medical imaging editorial</i>	Supplementary background; editorial overview of GDL in medical imaging.	Special issue overview across multiple topics.	Not a single empirical system; discusses themes and open challenges.	No directly comparable result set.	Useful for framing the broader medical GDL landscape.	Background only; not directly comparable as primary evidence.
S18 [18] Han et al. (2017) <i>High-resolution completion</i>	Supplementary methodological precursor; non-medical 3D shape completion.	General 3D object categories.	Global-structure inference + local patch refinement for high-resolution completion.	State-of-the-art performance reported in its original non-medical setting.	Useful mainly for tracing method lineage, not as direct medical evidence.	Historical method background; limited direct medical relevance.